METRIC

A-A-52423(AT)
January 25, 1993

COMMERCIAL ITEM DESCRIPTION

PUMP, SUBMERSIBLE, BILGE, AXIAL FLOW; ELECTRIC-MOTOR-DRIVEN

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-P-62156B(AT) which is canceled.

1.0 Abstract. This CID covers an electric-motor-driven, axial flow, submersible, bilge pump used to expel water and contaminations from the hulls of military, tactical and combat vehicles.

1.1 Configuration. The pump covered by this CID is available in five configurations. The part numbers and corresponding descriptions are listed below (see 5.2).

Part number	Description
8763300	With inlet elbow, without inlet screen.
8763300-1	Without inlet elbow, without inlet screen, with discharge elbow.
8763300-2	Without inlet elbow, without inlet screen, without discharge elbow.
8763300-3	Same as 8763300-2 except with a 90 degrees (°) electrical connector.
8763300-4	Same as 8763300-1 except with a 90° electrical connector.

2.0 Salient characteristics.

2.1 Materials. Materials used in the fabrication of the pump shall be as specified herein and in accordance with referenced standards, specifications and drawing. The use of recovered materials made in conformance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.4).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 4320

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2.2 Design and construction. The pump shall be designed and constructed as specified herein and in accordance with drawing 8763300 for the applicable part number (see 5.2). The pump shall operate in either a vertical (motor above the pump) or horizontal (with a 90° elbow) position. The pump shall be of such configuration as to permit the centerline of the propeller shaft to be within 6.3 centimeters (cm) of the mounting surface when installed in the horizontal position.

2.2.1 Weight. The weight of the pump shall be not greater than 5.4 kilograms (kg).

2.2.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. When dissimilar metals are used in intimate contact, suitable protection against galvanic corrosion shall be applied.

2.2.3 Threaded surfaces. The threads shall have a smooth finish and shall be free from flaws and other defects. The thread form, class and number of threads per inch shall be as specified on the applicable standards or drawing.

2.2.4 Electric motor. Unless otherwise specified, the motor shall be a direct current (de) series type and rated at 28 volt (V) for operation in an ambient temperature of 21 degrees Celsius (°C). The motor shall have provision for quick replacement of the brushes.

2.2.4.1 Brushes. The motor shall be provided with high grade brushes capable of operating continuously for not less than 500 hours (h) without replacement.

2.2.4.2 Power rating. The continuous-duty rating of the motor shall be not less than the brake horsepower requirement of the pump under any condition of head and capacity, from the rated output to free delivery.

2.2.4.3 Bearings. The motor shall be provided with anti-friction lubricated for life, sealed bearings, which shall function on a continuous-duty basis.

2.2.4.4 Temperature rise. The temperature rise of the motor, measured on the outside surface (base metal) of the housing over the poles when running in air at 25°C, shall be not greater than 71°C whether its pumping or running dry .

2.2.5 Interchangeability. To provide for interchangeability, all assemblies and components having the same part number shall be functionally and dimensionally interchangeable.

2.3 Performance.

2.3.1 Water level. The pump shall begin pumping with the water level as low as 2.5 cm above the mounting surface (inlet side) of the pump for vertical (shaft) installations, and for horizontal (shaft) installations, the pump shall begin pumping when the water level reaches the centerline of the propeller shaft.

2.3.2 Rated capacity. The pump shall deliver not less than 174 liters per minute (L/m) against a 1.8 meter (m) static pressure head through a 32 millimeter (mm) pipe outlet with shaft in the horizontal and vertical position and the motor at the stabilized operating temperature. The voltage at motor terminals shall be 28 Vdc and the specific gravity of water being pumped shall be not greater than 1.03.

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2.3.3 Dielectric strength. The pump shall show no evidence of puncture of solid insulation, continuing sparkover, breakdown of the dielectric material, or damage to the insulation sufficient to cause failure of the motor and shall meet the requirements of 2.3.2; after being exposed to a 500 V root-mean-square (rms) and 60 Hertz (Hz) potential for 1 minute.

2.3.4 Environmental ----- '

2.3.4.1 Temperature. The pump shall operate in a temperature range of O to plus (+) 52 degrees Celsius (°C).

2.3.4.2 Waterproofness. The pump shall evidence no moisture within the motor after operating, while submerged in a saline solution, for 30 minutes at full rated current and voltage with a minimum internal pressure of 40 kilopascals (kPh) applied to all voids.

2-3.4.3 Fungus resistance. The pump shall evidence no microbial growth that would adversely affect its performance and shall subsequently meet the requirements of 2.3.2 after exposure to fungus growth in air incubation cabinet for 90 days at a temperature between 24 and 30°C and a relative humidity between 93 and 100 percent.

2.3,4.4 Shock resistance. The pump shall evidence no permanently deformed, cracked, or broken parts and shall meet the requirements of 2.3.2 after being subjected to sawtooth waveform shock pulses of 30 gravity units (g) for 11 milliseconds (ins) time duration along each direction of the three mutually perpendicular axes (total of six shocks).

2.3.4.5 Vibration resistance. The pump shall evidence no permanently deformed, cracked, or broken parts and shall meet the requirements of 2.3.2 after exposure to single harmonic motion having an amplitude of 0.8 mm (1.6 mm maximum total excursion) through a frequency range of 10 to 55 Hz in each of the three perpendicular axes. The sweep time for the frequency range of 10 to 55 Hz and return to 10 Hz shall be 1 minute.

2.3.4.6 Electromagnetic interference. The pump shall not exhibit any malfunction or degradation of performance when subjected to electromagnetic Interference.

2.3.5 Endurance. The pump shall retain not less than 90% of the rated capacity specified in 2.3.2 after being operated for 350 hours.

2.4 Finish. Cleaning, pretreating, priming, and painting of exposed surfaces of the pump shall conform to an appropriate CARC paint system selected from Drawing no. 12369000.

2.5 Identification marking. Identification marking shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code, the contract number, the engineering drawing part number, and the national stock number (NSN) . Identification marking shall be accomplished without adversely affecting the acceptability and function of the finished product. Identification plates, if used, shall be photosensitive anodized aluminum 0.025 cm or thinner.

2.6 Workmanship. Workmanship shall be such as to assure a product free of cracked, broken, or rusted parts or components, scratched finish, peeled or blistered paint or any other defects which may affect its performance, serviceability, or appearance.

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3.0 Quality assurance provisions.

3.1 Responsibility for inspec-tion. The contractor is responsible for the performance of all inspections (examination and teats).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Government documents, drawings, and publications. Copies of Army drawings 12369000 "Chemical Agent Resistant Coating (CARC) Paint System Index" and 8763300 "Pump Assembly, Bilge" are available from the Contracting Officer, Us. Army Tank-Automotive Command, Warren, MI 48397-5000.

5.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Applicable drawing (see 2.2).
c. Applicable part number (see 1.1 and 2.2).
d. Issue of DODISS to be cited in the solicitation and if required the specific issue of individual documents referenced (see 5.1).
e. Selection of applicable level and packaging requirements (see 4.0).

5.3 Cross-reference data. Pumps conforming to this CID are interchangeable/ substitutable with pumps conforming to MIL-P-62156B(AT), dated 22 June 1989.

5.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Custodians:
Army - AT

Preparing activity:
Army - AT

(Project 4320-0365)

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